

PAVAN DHARMA ADAPA

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PROFESSIONAL SUMMARY

Data Analyst with an M.S. in Computer Science & Engineering, **available now for full-time roles**. Turns multi-source data into decisions using SQL, Python, Tableau, Power BI, and Advanced Excel. Experienced building end-to-end ETL pipelines, dimensional models, and executive dashboards, and rigorous about the statistics underneath them — complex-survey estimation, experiment design and sequential testing, model calibration, and variance reduction. Currently building and validating a research data pipeline at Mississippi State. Portfolio work runs on **real public datasets** at national scale (CDC, CMS, multi-hospital encounter data), with every figure reproducible from the published code.

TECHNICAL SKILLS

Data Analysis & Querying: SQL (joins, CTEs, window functions, query optimization), Python (Pandas, NumPy, SciPy, scikit-learn), R (ggplot2, dplyr), Advanced Excel (pivot tables, lookups, automation).

BI & Visualization: Microsoft Power BI, Tableau, Google Data Studio, Matplotlib, Seaborn — interactive dashboards, KPI reporting, executive-ready visualization.

Databases & Cloud: MySQL, PostgreSQL, SQLite, AWS; connecting BI tools to cloud-hosted sources.

Experimentation & Statistics: A/B test design, power and sample-size analysis, sequential testing (alpha spending, always-valid inference), CUPED variance reduction, sample ratio mismatch detection, hypothesis testing, complex-survey estimation, EDA.

Machine Learning: Logistic regression, Random Forest, XGBoost, gradient boosting, Gaussian Process Regression, MLP, cross-validation, model calibration, decision curve analysis.

Data Processing & Methodologies: ETL/ELT, data cleansing, wrangling, data mining, feature engineering, dimensional modeling, root cause analysis, KPI tracking, data pipelines.

Engineering Practice: Git, GitHub Actions CI, pytest, Snakemake, conda/Bioconda, Linux (WSL2), reproducible pipelines, hermetic test suites, technical documentation.

Other Tools: Linux, Selenium, NLP, LLMs, Jupyter.

PROFESSIONAL EXPERIENCE

Data Analytics Intern — Bioinformatics Pipeline

Jul 2026 – Present

Agricultural Microbiomes Lab, Mississippi State University — Starkville, MS

- Built and validated the pipeline the lab uses to turn Oxford Nanopore DNA sequencing output into bacterial abundance tables for poultry and catfish microbiome research.
- Ported the pipeline from the university's SLURM/HPC cluster to standard Windows and Linux workstations, removing every scheduler and cluster-storage dependency and cutting setup from a multi-day process to a single documented install.
- Cross-validated output against an independently developed implementation: the two agreed to **within 0.01 percentage points** on every species and genus, with identical read counts — the check that made the lab's results defensible.
- Processed and quality-controlled **2M+ sequencing reads** across 19 samples, and automated construction of the reference database from NCBI 16S RefSeq (27,700+ sequences, 21,000 taxa) into a single command.
- Root-caused an intermittent failure that destroyed results after an hour of compute, tracing it to host/guest clock drift rather than the cause the error message named; delivered the fix and the documentation.
- Contributed **6 merged pull requests** to the lab's open-source tool ahead of public release, covering packaging, failure diagnostics, citation metadata and contributor tooling.

Graduate Teaching Assistant

Sep 2025 – May 2026

Mississippi State University — Starkville, MS

- Led lab sessions in Python (Pandas, Matplotlib) and R (ggplot2, dplyr) across 6 data science courses, demonstrating how to clean, transform, and visualize datasets and communicate insight to non-technical audiences.
- Analyzed student performance data using Advanced Excel, SQL, and Python to track KPIs such as assignment completion, grade distribution, and retention, enabling instructors to identify at-risk students early.
- Designed and evaluated coding assignments simulating real analyst workflows — preprocessing, feature engineering, EDA, model evaluation — improving project completion speed by **25%**.

Graduate Research Analyst

Aug 2024 – Aug 2025

Mississippi State University — Starkville, MS

- Analyzed **900,000+** epidemiological records using Python, SQL, and EDA to identify key environmental drivers of disease transmission, producing data-backed recommendations for stakeholders in CSE and Veterinary Medicine.
- Built an automated ETL pipeline in Python and Selenium to collect, cleanse, and standardize multi-source datasets, reducing metadata processing time **95%** — from hours to under 10 minutes per dataset.
- Developed and evaluated ensemble predictive models (Gaussian Process Regression, Random Forest, XGBoost) to forecast disease outbreaks, achieving **87% predictive accuracy** and establishing baseline KPIs for model performance tracking.
- Automated feature-engineering and data-cleansing workflows, resolving **35%** of dataset inconsistencies and reducing manual analysis time **50%**.

Marketing Data Analyst Intern

May 2023 – Dec 2023

Viral Fission

- Developed digital marketing analytics and reporting systems tracking audience engagement, funnel performance, and campaign ROI for youth-focused brand campaigns with clients including Coca-Cola, ITC, and Mahindra.
- Built automated Power BI and GA4 dashboards that reduced manual reporting time **50%** and enabled real-time monitoring of reach, engagement rate, click-through rate, and conversion KPIs.
- Applied SEO optimization, A/B testing, and UX insight from GA4 and social analytics to refine content and posting strategy, contributing to a **25% increase** in key audience-interaction metrics.

Data Analytics Intern

Aug 2022 – Dec 2022

InfraBIM Techno Solutions

- Led analytics and reporting-automation initiatives on engineering and operations data using Power BI, SQL, and Google Colab, building KPI dashboards and ETL pipelines that cut manual reporting effort **~30%**.

ANALYTICS PROJECTS

Interactive Analytics Dashboard · Five-view BI application over every project's output

- Built a five-view analytics dashboard — clinical operations, revenue cycle, population health, risk model, experimentation — with **14 interactive charts, 21 KPI cards and drill-down tables**, covering every project below.
- Data layer is generated directly from each pipeline's output files rather than re-entered, so the dashboard **cannot drift** from the analyses it reports; charts are hand-written SVG with no third-party dependency.

Experimentation & A/B Test Analysis Toolkit · Estimator validation against known ground truth ·

github.com/adapapavandharma/experimentation-toolkit

- Quantified the peeking problem by simulation: stopping at the first significant look across 10 interim analyses drives the false positive rate to **18.5% against a nominal 5%**; implemented O'Brien–Fleming alpha spending and always-valid mSPRT bounds and measured what each costs in power.
- Implemented **CUPED** variance reduction matching the p^2 prediction to within a tenth of a point — at $p = 0.7$ worth **1.96× the traffic at no cost** — with unbiasedness verified across 300 replications.
- Built sample-ratio-mismatch detection and demonstrated a global SRM check **passing cleanly** on an experiment whose segments are catastrophically imbalanced, establishing per-segment SRM as the necessary control.

Claims Denials & Appeals Analytics · CMS Transparency in Coverage — 3-year national panel · github.com/adapapavandharma/denials-analytics

- Built a 15,545 plan-year panel covering **126.9 million denied claims** across every Qualified Health Plan on the federal Marketplace, 2021–2023.
- Established that only **0.168% of denials are ever appealed** while **40.1% of appeals succeed**, identifying capacity to file — not claim merit — as the binding constraint on denial recovery.
- Showed the payer-reported reason taxonomy cannot support root-cause work ("**Other**" is **71.7%** of categorised denials and growing), and that denial rate is an organisational property: **6.0% to 48.3%** across issuers with 100,000+ claims.

Emergency Department Throughput Analytics · CDC/NCHS NHAMCS — real national survey microdata ·

github.com/adapapavandharma/ed-throughput-analytics

- Analyzed **16,025 real emergency department visits**, survey-weighted to the **155.4M** US ED visits of 2022; the pipeline self-validates by reproducing CDC's published national total exactly or refusing to run.
- Implemented the **ultimate-cluster variance estimator with Taylor linearisation** CDC prescribes for this stratified multi-stage sample; design-correct inference showed the hypothesised peak-hour effect does **not** clear significance at $\alpha = 0.01$ — a negative result reported rather than discarded.
- Found between-hospital variation in median door-to-provider time (2–74 min) to be **3.9× larger** than time-of-day variation, and quantified boarding at **48.6M ED bay-hours lost annually** as the recoverable capacity item.

30-Day Readmission Risk & Care-Management Registry · 99,343 real inpatient encounters, 130 hospitals ·

github.com/adapapavandharma/readmission-risk

- Built a gradient-boosted readmission model (**AUC 0.664**) on real multi-hospital encounter data; top-quintile targeting captures **37.7% of all readmissions** at **1.89× enrichment**, number-needed-to-screen 4.8, delivered as a capacity-derived tiered risk board.
- Benchmarked against a single-variable clinical rule to isolate marginal value — prior inpatient count alone reaches 32.2% recall, so the model buys **+5.5 points**, materially less than an AUC comparison implies.
- Showed balanced class weighting inflates mean predicted risk to **46% against an observed 11%** while leaving AUC unchanged, establishing calibration rather than discrimination as the constraint on deployability.

EDUCATION

M.S., Computer Science & Engineering

Graduated May 2026

Mississippi State University — Mississippi State, MS

Coursework: Advanced Machine Learning, Artificial Intelligence, Algorithms, Parallel Algorithms, Software Engineering, Data Analytics, Advanced Cyber Operations.

B.Tech, Computer Science & Engineering

Graduated May 2024

Malla Reddy University — Hyderabad, India

PUBLICATION (IN PROGRESS)

Computational Tools for Modeling and Predicting Respiratory Disease Spread in Commercial Poultry Production

Zhang Li, Pavan Dharma Adapa, Sai Harika Gade, Saikanth Ratnavale, Navicky Michael — compartmental (SIR/SEIR) and machine-learning forecasting of respiratory disease transmission at scale.